

Module 16: Row-Level Security (RLS) & Data Security

Overview

Data is one of an organization's most valuable assets. While dashboards provide insights, not every user should have access to all data.

A Sales Manager may need access to all regions, while a Sales Representative should only see data for their assigned region.

Power BI provides Row-Level Security (RLS) and other security features that enable organizations to control data visibility, enforce governance, and protect sensitive information.

This module covers Power BI security fundamentals, Static RLS, Dynamic RLS, security roles, `USERPRINCIPALNAME()`, Object-Level Security (OLS), and enterprise security best practices.

Learning Objectives

After completing this module, you will be able to:

- Understand Power BI security concepts
 - Implement Row-Level Security (RLS)
 - Create Static RLS roles
 - Create Dynamic RLS solutions
 - Use `USERPRINCIPALNAME()`
 - Configure security roles
 - Test security implementations
 - Understand Object-Level Security (OLS)
 - Apply enterprise security best practices
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1 Introduction to Data Security

What is Data Security?

Data Security refers to protecting data from unauthorized access, misuse, or exposure.

Security Objectives

Confidentiality

Only authorized users can access data.

Integrity

Data remains accurate and unchanged.

Availability

Authorized users can access data when needed.

CIA Triad

```
Confidentiality
  +
Integrity
  +
Availability
```

2 Why Security Matters

Organizations often store:

- Financial Data
 - Employee Data
 - Customer Data
 - Sales Data
 - Strategic Information
-

Example

A sales employee should not view:

- Executive salary information
 - Other departments' confidential data
 - Restricted financial reports
-

Consequences of Poor Security

- Data Leaks
 - Compliance Violations
 - Financial Losses
 - Reputational Damage
-

Security Layers in Power BI

Power BI security exists at multiple levels.

Layer 1

Workspace Access

Controls who can access content.

Layer 2

App Permissions

Controls who can consume reports.

Layer 3

Row-Level Security

Controls which rows users can see.

Layer 4

Object-Level Security

Controls which tables or columns users can see.

Layer 5

Data Source Security

Database-level protection.



What is Row-Level Security (RLS)?

Definition

Row-Level Security (RLS) restricts data visibility at the row level.

Example

Sales Table

Region	Revenue
North	₹100,000
South	₹150,000

User A

Can see:

Region	Revenue
North	₹100,000

User B

Can see:

Region	Revenue
South	₹150,000

Same Report

Different users see different data.

5 Types of RLS

Power BI supports two primary approaches.

Static RLS

Rules are manually defined.

Dynamic RLS

Rules are determined automatically based on the logged-in user.

6 Static RLS

What is Static RLS?

Static RLS uses fixed filters.

Example

Role:

```
North Region
```

Filter

```
Sales[Region] = "North"
```

Result

Users assigned to this role see only North region data.

Advantages

- Easy to implement
 - Simple maintenance
-

Disadvantages

- Difficult to scale
- Requires many roles

7 Dynamic RLS

What is Dynamic RLS?

Dynamic RLS automatically determines what data users can access.

Example

Email	Region
north@company.com	North
south@company.com	South

Users see data based on their email address.

Benefits

Scalable

Supports thousands of users.

Easier Maintenance

Single security model.

Enterprise Standard

Most organizations use Dynamic RLS.

8 USERPRINCIPALNAME()

What is USERPRINCIPALNAME()?

Returns the email address of the currently logged-in user.

Example

```
USERPRINCIPALNAME()
```

Possible Result

```
john@company.com
```

Why Important?

Foundation of Dynamic RLS.



Security Mapping Tables

Dynamic RLS usually requires a mapping table.

Example

Security Table

Email	Region
john@company.com	North
sara@company.com	South

Relationship

```
Security Table  
  ↓  
Sales Table
```

Dynamic Filter

```
Security[Email] =  
USERPRINCIPALNAME()
```

Result

Users automatically see authorized data.

Creating Security Roles

Step 1

Open Power BI Desktop.

Step 2

Navigate to:

```
Modeling
↓
Manage Roles
```

Step 3

Create Role.

Example:

```
North Region
```

Step 4

Define Filter.

```
Sales[Region] = "North"
```

Step 5

Save Role.

Testing RLS

Why Test?

Verify users only see authorized data.

Steps

Modeling



View As

Example

Test:

North Region

Power BI simulates the selected role.

Best Practice

Test every role before deployment.

1 **2** Publishing RLS Models

After publishing:

Step 1

Open Power BI Service.

Step 2

Navigate to:

Semantic Model

Step 3

Select:

Security

Step 4

Assign users to roles.

Example

North Region

↓

john@company.com

Result

Role becomes active.

1 **3** Object-Level Security (OLS)

What is OLS?

Object-Level Security restricts access to:

- Tables
 - Columns
-

Difference

RLS

Controls rows.

OLS

Controls objects.

Example

Finance Team:

Can view:

Salary

Sales Team:

Cannot view:

Salary

Use Cases

HR Systems

Payroll Reports

Financial Data



Workspace Security

Workspace permissions control content access.

Roles

Admin

Full control.

Member

Can edit content.

Contributor

Can publish content.

Viewer

Read-only access.

Important

Workspace permissions do not replace RLS.

Data Governance

What is Governance?

Governance ensures secure and controlled data usage.

Goals

Consistency

Single source of truth.

Security

Protect sensitive information.

Compliance

Meet regulations.

Accountability

Track usage and access.

Enterprise Security Best Practices

Use Dynamic RLS

Preferred for scalability.

Implement Least Privilege

Grant only required access.

Test Security Thoroughly

Before deployment.

Document Security Roles

Maintain role definitions.

Monitor Access

Audit user activity.

Use Security Groups

Manage users through groups.

Protect Sensitive Columns

Use OLS where needed.

Common Security Mistakes

No RLS Implementation

Exposes all data.

Excessive Permissions

Violates least privilege principle.

Hardcoded Security Logic

Difficult to maintain.

Not Testing Roles

May expose unauthorized data.

Ignoring Governance

Creates compliance risks.

Key Terminologies

Term	Definition
RLS	Row-Level Security
OLS	Object-Level Security
Static RLS	Fixed security filters
Dynamic RLS	User-based security
USERPRINCIPALNAME()	Returns current user email
Security Role	Collection of security rules
Security Table	Maps users to permissions
Governance	Controlled data management
Least Privilege	Minimum required access
Compliance	Regulatory adherence

 Practice Exercises

Exercise 1

Create a Static RLS role:

```
Sales[Region] = "North"
```

Exercise 2

Test RLS using:

```
View As
```

Exercise 3

Create a Security Mapping Table.

Exercise 4

Implement Dynamic RLS using:

```
USERPRINCIPALNAME()
```

Exercise 5

Compare:

- RLS
- OLS

 Mini Project

Secure Sales Analytics Dashboard

Dataset

Sales Data

Region	Revenue
--------	---------

Security Table

Email	Region
-------	--------

Requirements

Dynamic RLS

Users see only assigned region.

Roles

- Sales Manager
- Sales Representative

Testing

Verify multiple users.

Deliverables

1. Security Mapping Table

2. Dynamic RLS Configuration
3. Role Documentation
4. Security Testing Results



Module Summary

In this module, you learned:

- ✓ Data Security Fundamentals
- ✓ Power BI Security Layers
- ✓ Row-Level Security (RLS)
- ✓ Static RLS
- ✓ Dynamic RLS
- ✓ USERPRINCIPALNAME()
- ✓ Security Mapping Tables
- ✓ Role Management
- ✓ Testing Security
- ✓ Object-Level Security (OLS)
- ✓ Workspace Security
- ✓ Data Governance
- ✓ Enterprise Security Best Practices

Security is a critical component of enterprise Power BI solutions. Proper implementation of RLS and governance ensures users access only the data they are authorized to see while maintaining compliance and protecting sensitive business information.



Next Module

Module 17: Power BI Performance Optimization

Topics Covered:

- Performance Fundamentals
- Performance Analyzer
- Query Optimization
- DAX Optimization
- Data Model Optimization
- Storage Optimization

- Star Schema Performance
- Incremental Refresh
- Aggregations
- Enterprise Performance Best Practices